Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_14

1. Summary of Key Deviations

The analysis of Run_14 indicates a very minor, localized anomaly. The vast majority of the system is operating perfectly.

- **Flow_Bypass (Moderate):** The only significant anomaly is a consistent, moderate positive **offset** on the `Flow_Bypass` signal, which reads approximately **+9% to +10%** higher than healthy across all active states (s2–s7).
- **Flow_Pump (Minor):** A corresponding, but much smaller, minor positive **offset** of around **+4%** is visible on the `Flow_Pump` signal.
- **All Other Signals (Nominal):** Crucially, there are **no deviations** on any other signals. All pressures (`p_Pump`, `p_FMU`, `p_Combustion`, `p_Shut`), all engine flows (`Flow_Engine1-4`), and the high-pressure relief flow (`Flow_PRV`) are operating perfectly within their healthy ranges.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Flow_Bypass Sensor**	**Calibration Shift (Positive Offset)**	This is the most probable cause. The `Flow_Bypass` sensor is the only one showing a moderate deviation (+10%), and this deviation occurs in complete isolation from any physically expected consequences. If 10% extra flow were truly being bypassed, there would be a noticeable drop in downstream pressures and engine flows, none of which are observed. This points to a sensor simply *reporting* a high value, not a real physical event.

2	**Bypass Valve**	**Minor Internal Leak**	It's possible the bypass valve has a small internal leak, allowing a small amount of extra flow to pass through that is detected by the `Flow_Bypass` sensor. This would be a real physical fault. However, given that no other part of the system seems affected, a sensor calibration error is a more direct and simpler explanation.
3	**Flow_Pump Sensor**	**Minor Calibration Shift**	The `Flow_Pump` sensor also shows a minor +4% offset. It's possible both this sensor and the `Flow_Bypass` sensor have minor, independent calibration errors.
4	**Main Fuel Pump**	**Minor Over-performance**	The pump could be delivering ~4% more flow than commanded. This would explain the `Flow_Pump` reading and could contribute to a slightly higher `Flow_Bypass` reading. However, the `Flow_Bypass` deviation is much larger (+10%), suggesting the issue is more localized to that sensor.
5	**Wiring / Electrical Fault**	**Signal Noise / Grounding Issue**	A minor electrical issue in the `Flow_Bypass` sensor's circuit could introduce a positive bias or offset to the signal, explaining the isolated reading without any real change in the physical system.

3. Most Probable Hypothesis

The most probable root cause is a **Calibration Shift creating a Positive Offset on the Flow_Bypass Sensor.**

Reasoning: This is a classic case of an isolated sensor anomaly. The `Flow_Bypass` signal is the only one showing a moderate deviation. The complete absence of any secondary effects—no pressure changes, no engine flow changes—is the critical piece of evidence. A real physical diversion of 10% of the system's flow would have significant and measurable consequences throughout the entire fuel line. Since none exist, the most logical and parsimonious conclusion is that the physical system is healthy and only the `Flow_Bypass` sensor is providing an incorrect, artificially high reading.